

2

The output of this phase is an E-R diagram 

- ☒ Conceptual Design ✓
- ☐ Logical Design
- ☐ Physical Design

✗ **Incorrect** 0/1 Points

3

Merge local models into global model 

- ☒ Conceptual Design
- ☐ Logical Design ✓
- ☐ Physical Design

6

Design user views 

- ☐ Conceptual Design
- ☐ Logical Design
- ☒ Physical Design ✓

✓ **Correct** 1/1 Points

7

Define integrity constraints 

- ☐ Conceptual Design
- ☒ Logical Design ✓
- ☐ Physical Design

relation the combination of the two (so "AE") 

A, B, C, XD, E

Correct answers:

XD becomes one entity. Others are just what their names are.

✓ **Correct** 2/2 Points

35

If the cardinality for each relationship is changed to one-to-one with mandatory participation for all entities; how many relations would be derived from this version of the ER model? 

1

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4

Identify relationship types 

- ☒ Conceptual Design ✓
- ☐ Logical Design
- ☐ Physical Design

✓ **Correct** 1/1 Points

5

Determine key (CK, PK) attributes 

- ☒ Conceptual Design ✓
- ☐ Logical Design
- ☐ Physical Design

10

Design security mechanisms 

- ☐ Conceptual Design
- ☐ Logical Design
- ☒ Physical Design ✓

✓ **Correct** 1/1 Points

11

Choose indexes 

- ☐ Conceptual Design
- ☐ Logical Design
- ☒ Physical Design ✓

8

Review with the users 

- ☒ Conceptual Design ✓
- ☒ Logical Design ✓
- ☒ Physical Design ✓

✓ **Correct** 1/1 Points

9

Normalization 

- ☐ Conceptual Design
- ☒ Logical Design ✓
- ☐ Physical Design

14

What is an entity, and how is it related to a table? Explain why an entity and a table are not exactly the same. 

An entity represents a real-world object or concept, while a table is representation of that entity. Entities exist

Correct answers:

Entity is a representation of a concept. Many entities become tables, but some tables represent relationships between entities or multi-valued attributes. Also, some entities merge to become a single table.

✓ **Correct** 6/6 Points

15

What does SRAM stand for and how is it used? 

This question is not related to this course.

Correct answers:

This question is not related to this course

12

Long Answer Section - 5 questions (13 - 17)

In this section, there are two types of questions:

1. Questions asking you to explain a concept from this course material. You must respond with an answer between two (2) and four (4) sentences. Anything substantially longer than this will receive 0 points.
2. Questions that are not related to this course subject matter. You must respond "This question is not related to this course". Any other response, even if it is correct will receive 0 points. A blank response will receive 0 points.

NOTE: If you receive 0 points on three (3) or more of these questions, your exam will be reviewed in detail for suspicious responses and might receive 0 points for the entire exam. 

✗ **Incorrect** 1/6 Points

13

Which task that is part of the LOGICAL design process is considered optional? Explain why this task is optional. 

In the Logical design process, review with user is optional task because might not be available to meet even

Correct answers:

Merge into global database. The design's DB could be small enough that not other DB to merge with

17

What do HTML and XML stand for? Give one example keyword from each and explain how the keyword is used. 

This question is not related to this course.

Correct answers: This question is not related to this course

✖ Incorrect

18

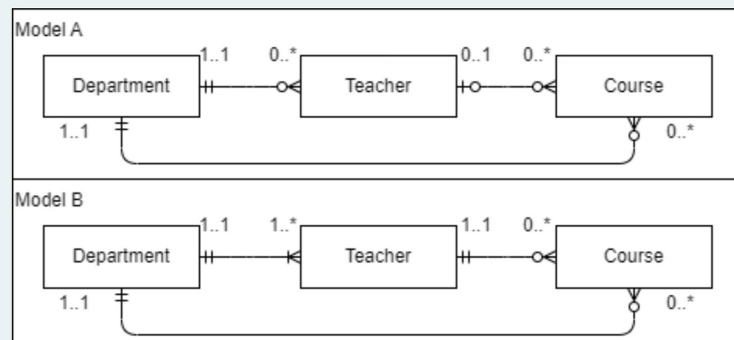
Normalization Section - 8 questions (22 - 29)

Your team is tasked with designing the Student scheduling application for Courses in a university. Below are some example screens from the interface for a student. You will be asked questions based on how you would normalize a database for this system. NOTE: for this section, you are only designing the portion of the system that students will use. 

✖ Incorrect

16

Given the two shown ER diagrams, one has resulted from fixing a trap and the other has a redundant relationship. What is the type of trap that was fixed? Identify which diagram is which and how do you know? 



Course and Department relations is redundant. Model B is result because in the Model A, a course can have

Correct answers: Model A fixes a chasm trap - the optional participation on the Teacher side. Model B has a redundant relationship.

✖ Incorrect

20

Screen 2: Look Up Classes

Fall 2023		
Department: Computer Engineering		
1000	Personal Computer Applications	View Sections
1200	Intro to Computers	View Sections
1210	Fundamentals of Computing I	View Sections

✖ Incorrect

21

Screen 3: Sections

✖ Incorrect

19

Screen 1: Grade Summary for Spring 2021 for Student Nusret Gökçe

Dept	Course	Section	Title	Grade	Credit	Earned	GPA points
COMP	3000	3	Object Oriented Programming	AA	3	3	12
COMP	3270	2	Intro to Algorithms	DD	3	3	3
COMP	3350	1	Computer Organization and Assembly	BB	3	3	9
COMP	3500	1	Intro to Operating Systems	FF	4	0	0
GPA Summary							
	Attempted	Earned	GPA Points	GPA			
Current Term	13	9	24	1.85			
Overall	80	71	184	2.30			

✖ Incorrect

20

Screen 2: Look Up Classes

✖ Incorrect 8/12 Points

22

Explain what tables you would create and for each, what would be the PK and FKs.

I create Departments, Courses, Sections, Instructors, Students, and Grade Tables.
Departments table have DepartmentID (PK), DepartmentName varchar(255), and CourseID (FK)
Courses table have CourseID (PK), DepartmentID (FK), InstructorID (FK), SectionID(FK), GradeID(FK), and also CourseTitle varchar(255).
Sections table have CourseID (FK), SectionID (PK = Combine them (CourseID and SecitonID)), _Days varchar(2), _Time Time, capacity int, actual int, Credit int.
Instructors table have InstructorID (PK) InstructorName varchar(255), CourseID(FK)
Students table have StudentID(PK), CourseID(FK), and StudentName varchar(255).
Grade table have GradeID (PK), GradeName varchar(2), StudentID (FK), SectionID(FK), Earned int, GPA int.

✔ Correct 4/4 Points

23

Identify one example of derived data from the screens shown. Explain what the source data is and why the item you listed is considered derived.

✖ Incorrect

21

Screen 3: Sections

Add	Dept	Course	Section	Credit	Title	Days	Time	Capacity	Actual	Rem	Instructor
☐	COMP	1210	1	3	Fundamentals of Computing I	MW	12:00-13:15	25	25	0	John Doe
☐	COMP	1210	2	3	Fundamentals of Computing I	MW	13:30-14:45	25	25	0	John Doe
☐	COMP	1210	3	3	Fundamentals of Computing I	MW	15:00-16:15	25	21	4	John Doe
☒	COMP	1210	4	3	Fundamentals of Computing I	MW	16:30-17:45	25	13	12	John Doe

✖ Incorrect 8/12 Points

22

Explain what tables you would create and for each, what would be the PK and FKs.

✖ Incorrect 3/5 Points

25

Explain a possible request as user could ask for that would display data from multiple tables that are in the system described in this section. Below that, type the SQL query you would write to perform the request.

```
SELECT d.DepartmentName, c.CoursName
From Department d
LEFT JOIN Course c ON c.DepartmentID = d.DepartmentID AND c.CourseID= d.CourseID
```

✔ Correct 5/5 Points

26

Write a possible request as user could ask for that would display data involving an aggregate function from the tables in the system described in this section. Below that, type the SQL query you would write to perform the request.

```
SELECT s.StudentID, c.CourseName, SUM(g.Grade) / SUM(c.Credit) AS AverageGradePoint
FROM Students s
JOIN Course c ON s.CourseID = c.CourseID
JOIN Grade g ON c.GradeID = g.GradeID
GROUP BY s.StudentID, c.CourseName;
```

✔ Correct 4/4 Points

23

Identify one example of derived data from the screens shown. Explain what the source data is and why the item you listed is considered derived.

We can derive the data GPA. GPA is equal to SUM(GPA)/SUM(Credit).

Correct answers:

Actual, Rem, Grade (technically), Earned, GPA Points, Attempted, Earned, GPA Points, GPA

✔ Correct 6/6 Points

24

We discussed two ways to represent derived data. For the item you wrote above, which of the two ways would you choose to represent the data and explain why.

I choose tables to represent derived data because it needs to update when I add new data and also it must be calculated every time, for example, GPA. When I add new data it needs to be calculated. If I use views, I need to create views again to see new data in the view.

✗ Incorrect 3/5 Points

29

Write a possible request as user could ask for that would remove data from a table in the system described in this section. Below that, type the SQL query you would write to perform the request.

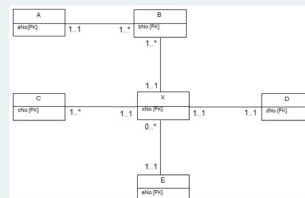
Delete Course WHERE CourseID = 2

✗ Incorrect

30

Mapping Section - 5 questions (31 - 35)

Answer the following questions with reference to how the ER model in the figure maps to relational tables.



✗ Incorrect 3/5 Points

27

Write a possible request as user could ask for that would add data to a table in the system described in this section. Below that, type the SQL query you would write to perform the request.

INSERT INTO Instructors(InstructorID, InstructorName, CourseID)
VALUES (2, 'Ibrahim Duman', 4)

✗ Incorrect 3/5 Points

28

Write a possible request as user could ask for that would modify data in a table in the system described in this section. Below that, type the SQL query you would write to perform the request.

Update Sections
Set Days= TW, Time = 13:30-14:45
Where SectionID = 1

✓ Correct 2/2 Points

33

Which relation(s) will have no foreign key?

B,C

Correct answers: B, C

✓ Correct 5/5 Points

34

Using only the letter identifier for each entity, provide appropriate names for the relations mapped from the ER model. (i.e. if an entity (such as A) becomes one relation, name that relation the same as the entity (so "A"). if two entities (like perhaps A and E) become one relation, name that relation the combination of the two (so "AE")

A, B, C, XD, E

Correct answers: XD becomes one entity. Others are just what their names are.

✓ Correct 2/2 Points

31

How many relations will represent the ER model?

5

✗ Incorrect 0/2 Points

32

How many foreign keys will be in the relation representing X? Which relation(s) will they reference?

2 - B and C

Correct answers: 1 - E (possibly 1 to D, but that is not necessary)

✓ Correct 2/2 Points